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CHAPTER 3

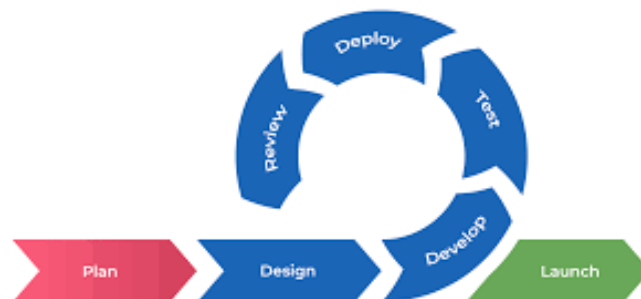
METHODOLOGY

Research Approach

This study employs the Agile software development methodology to guide the creation of the Interactive Learning Management System (iLearnSystem) for Computer System Servicing (CSS) students at Paliparan III Senior High School. Agile was selected because the requirements of a learning management system are not fully known at the start; they emerge as teachers and students interact with early versions of the platform. The methodology allows the research team to respond to user feedback quickly without discarding previous work.

Figure 1

Agile Methodology Diagram



Agile Methodology Diagram starts with the **Plan** phase, which defines the project's scope, goals, and timelines, where researchers identify potential clients and



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carry out preliminary interviews to evaluate their requirements. The **Design** phase is where researchers iteratively design the system architecture, user interface and user experience, and data models based on the system's identified needs. In the **Develop** phase, researchers begin the actual coding and create the system itself based on the finalized design from the previous phase. The **Testing** phase is where researchers conduct tests and evaluations to ensure the system's integrity and that it delivers exactly what it was built to do, identifying any defects or missing functionalities before release. The **Deployment** phase is the critical stage where tested, high-quality software increments are released into a live production environment for end-users to access and use. During the **Review** phase, researchers hold a meeting with stakeholders to demonstrate the system, gather feedback, and discuss any improvements or changes needed for future iterations. Lastly, the **Launch** phase is the formal, strategic introduction of the new product, major feature, or update to the market and users, often involving marketing, sales, and support coordination to ensure successful adoption. The entire process is iterative, meaning that after the Review phase, the team may return to the Plan or Design phase for the next cycle, allowing continuous improvement based on stakeholder feedback. This cyclical nature makes Agile particularly effective for projects where requirements may evolve over time, as it enables teams to respond quickly to changes without waiting for a final, monolithic release. Each iteration produces a

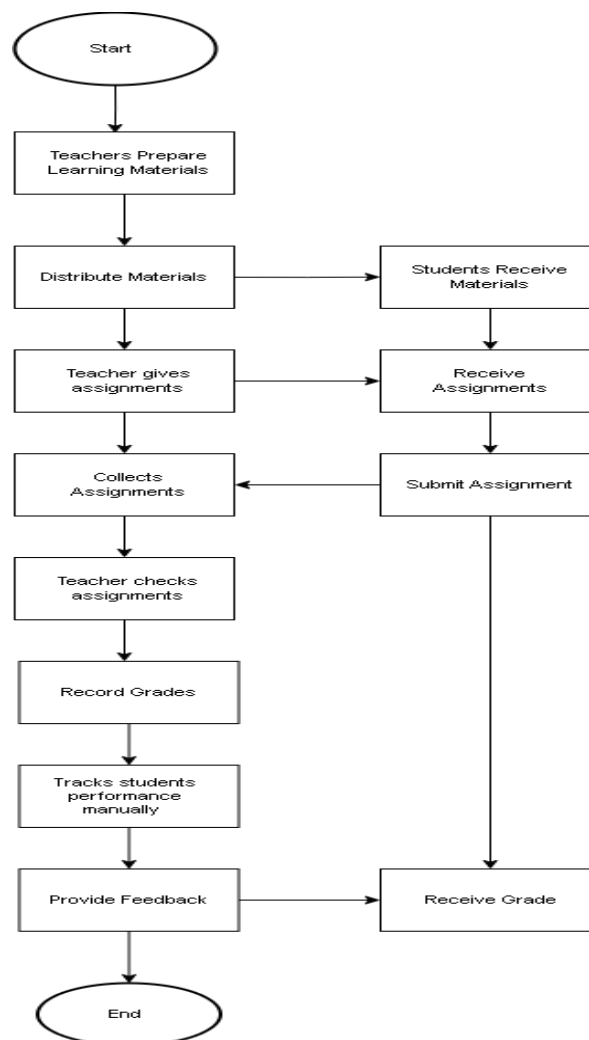


working increment of the software, ensuring that value is delivered to users early and often throughout the project timeline teams.

Requirement Analysis

Figure 2

ILearn Current Process





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Figure 2 The process begins with teachers preparing learning materials by gathering or creating lesson content, after which they Distribute Materials to the students through printed copies or messaging apps, and then the Students Receive Materials and keep them for study. Next, the Teacher gives assignments related to the lesson, and the students receive Assignments by noting down the instructions and deadlines. After working on the tasks, the students submit assignments by passing printed papers or sending files, and then the teacher collects assignments manually using a checklist. Following collection, the Teacher checks assignments by grading each submission one by one, then Record Grades into a grade book or spreadsheet, and then the teacher tracks students' performance manually by calculating averages and identifying struggling students without any automated tool. After that, the teacher provides feedback to students through written or verbal comments, and the students receive Grade either on their returned paper or through a message.



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Figure 3

Use Case Diagram

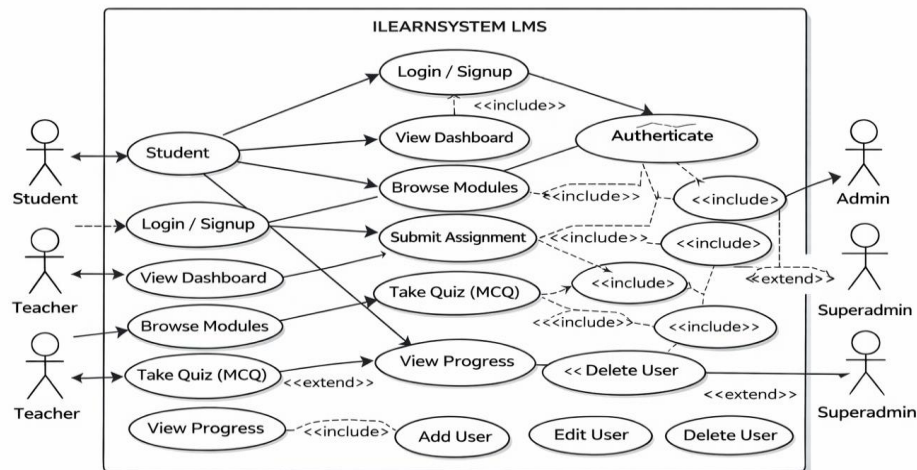


Figure 3 shows the use case diagram for the Interactive Learning Management System. The diagram has four user types, they are the Super Admin, Admin, Teacher, and Student. Super Admin holds the highest level of authority within the system. It is responsible for full control over the entire platform. This includes managing all user accounts across the institution (students, teachers, and admins), assigning roles, and resetting passwords. The Super Admin can configure system-wide settings such as authentication methods, session timeouts, data encryption parameters, and audit log retention. Admin (or Administrator) serves as an intermediate manager who supports daily operations but does not have full system control. Responsible for user management within a limited scope—such as creating,



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editing, or deleting teacher and student accounts for a specific department or grade level. They can also oversee course listings, generate basic reports on system activity, and perform routine maintenance tasks like data backups. However, the Admin does not have access to global system settings or the ability to modify security configurations. Teachers are responsible for creating and organizing classes, uploading learning materials, designing assessments, grading submissions, and monitoring student performance. They can upload modules and organize them by grade level and section, attaching files (PDFs, PowerPoints, Word documents) and embedding instructional videos. They can create quizzes and assignments with various question types, set due dates, and assign point values. Teachers have the ability to post announcements to the entire class or individual students. For evaluation, teachers can view all submitted assignments, grade submissions manually or through automatic scoring (for objective quizzes), and record grades within the system. Allows teachers to monitor module completion rates, quiz results, and overall student performance in real time through visual dashboards and reports. Teachers can provide written feedback on submissions and communicate with students via the integrated messaging system. The Student is the primary learner in the system. They are responsible for accessing and engaging with learning materials, completing assessments, and monitoring their own academic progress. Specifically, a student can log in and view a personalized dashboard that displays enrolled subjects, pending tasks, module completion progress, and upcoming



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deadlines. Through the system, students can browse interactive learning modules that include multimedia content such as videos, flashcards, and quizzes. Students can take quizzes, submit assignments in PDF or other formats, and view automated feedback and grades. Students can receive announcements and notifications from teachers, view their own performance trends through visual progress reports, and communicate with teachers via messaging or comment features.

Figure 4

Story Board

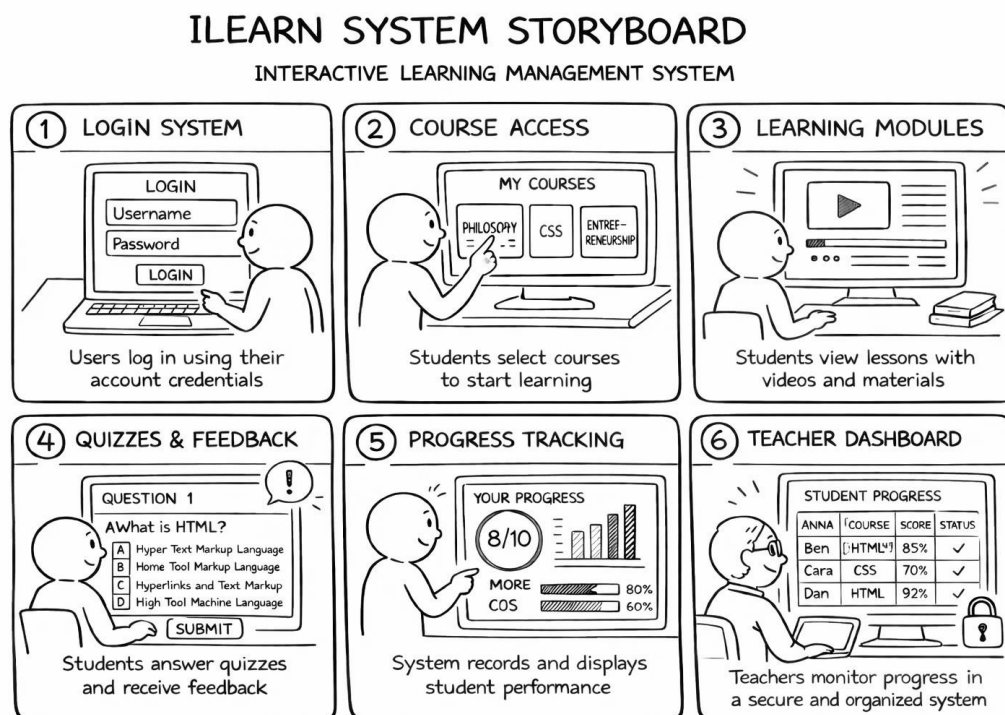


Figure 4 presents the storyboard for the Interactive Learning Management System (ILEARN SYSTEM) for CSS Students at Paliparan III Senior High School,



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showing the key user interfaces and workflows across six panels that address the current process and showcase the proposed solution. The **first** depicts the Login System, where users (students, teachers, administrators and superadmin) enter their username and password to access the platform. This panel addresses the lack of secure authentication in the current manual setup, ensuring that only authorized individuals can enter the system and that role- based access is enforced as specified in the research objectives on user authentication and security. The **second** panel shows Course Access, labeled as "MY COURSES" with subject options such as Math, Science, and History. In the context of the CSS strand, this panel represents the student's dashboard where enrolled subjects (e.g., Introduction to Philosophy, CSS, Entrepreneurship) are displayed. Students can select a course to begin learning, solving the problem of fragmented access to learning materials that previously required searching through multiple platforms or printed handouts. Next is the **third** panel illustrates the Learning Modules interface, where students view lessons that include videos and other multimedia materials. This panel directly implements the interactive learning module feature which incorporates instructional videos, flashcards, and page- by- page content delivery. The panel demonstrates how students can engage with rich media content within a single, centralized platform, replacing the previous method of passively receiving printed or static digital files. The **fourth** panel presents Quizzes & Feedback, showing a sample multiple- choice question ("What is HTML?") with four answer options and a submit button. This panel



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addresses the assessment and evaluation objective, allowing students to take quizzes and receive immediate feedback. The system automatically scores objective question types (multiple- choice, true/false, multi- select), providing instant results to students and reducing the teacher's manual grading workload as described in the automated progress tracking feature. The **fifth** panel displays Progress Tracking, with a visual indicator showing "8/10" and "80% / 60%" completion or performance metrics. This panel demonstrates the automated progress tracking tool that evaluates student performance based on quiz results, module completion, and assignment submissions. The system generates visual progress reports and dashboards for both students and teachers, promoting transparency and a deeper understanding of academic progress as outlined in the study's objectives. And lastly the **sixth** panel shows the Teacher Dashboard, where teachers can monitor student progress with a secure and organized interface. The panel lists student names (Anna, Ben, Cara), their courses, and their scores (85%, 70%, 92%). This panel addresses the teacher's need to efficiently monitor module completion, quiz results, and overall performance in real time, as well as to identify students who may need academic support. The dashboard replaces the manual method of tracking student performance using spreadsheets or paper grade books, enabling timely feedback and data- driven interventions.

Figure 5

Context Diagram



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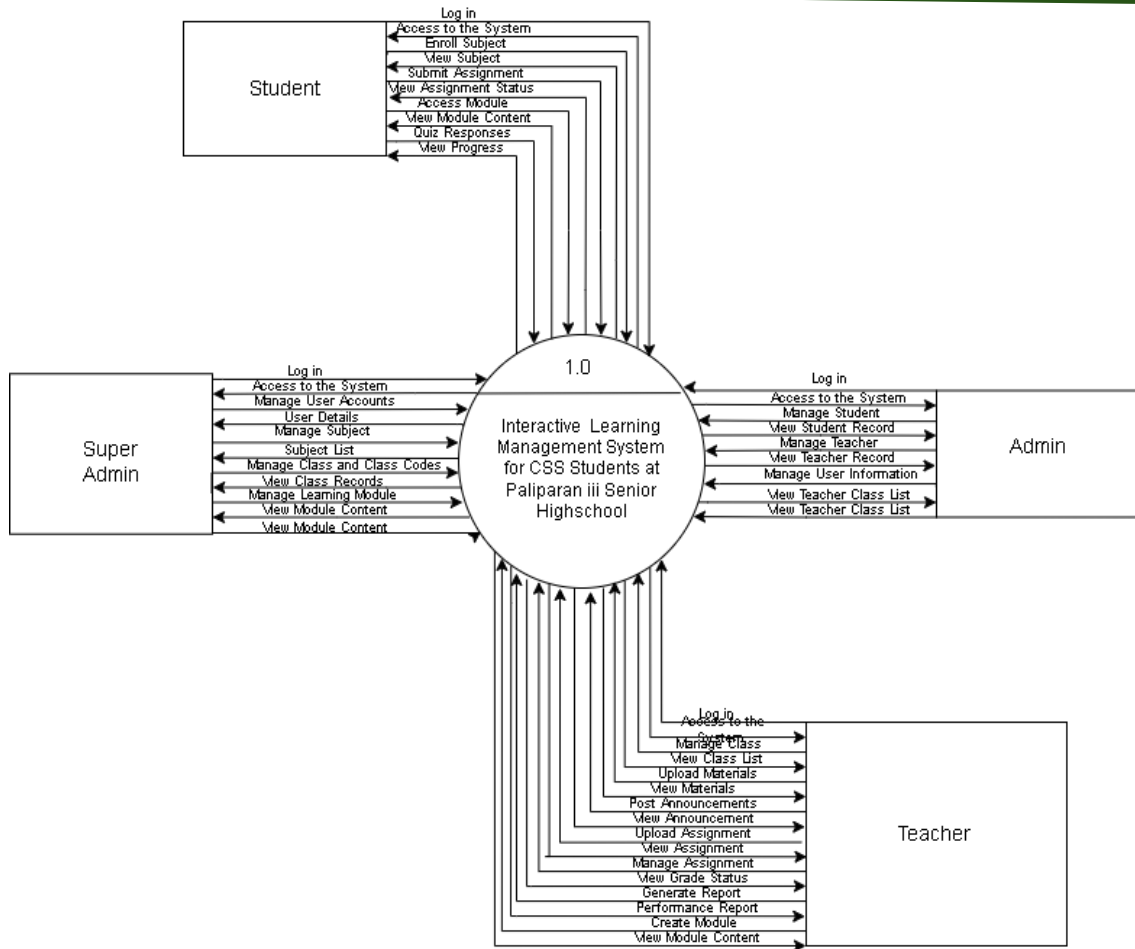


Figure 5 shows the context diagram of the Interactive Learning Management System for Computer System Servicing (CSS) Students at Paliparan III Senior High School, which outlines the high-level interaction between external entities: the Student, Teacher, Admin, and Super Admin. The Super Admin holds the highest level of authority within the system, with capabilities that include managing all user accounts (students, teachers, and admins), configuring system-wide settings such as authentication and security parameters, generating overall system performance reports and audit logs, and performing maintenance tasks like database backups



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and software updates. The Admin serves as an intermediate manager, responsible for creating, editing, and deleting teacher and student accounts within a limited scope (e.g., a specific department or grade level), generating basic usage reports, and handling routine system maintenance such as data backups, but without access to global security configurations or full system settings.

On the other hand, the Teacher is the content manager and academic facilitator, who can upload learning modules, create quizzes and assignments, post announcements, grade student submissions, view student progress through automated analytics dashboards, and communicate with students via messaging features – all mediated through the system. The Student, who is the primary learner, can access interactive learning modules (including videos, flashcards, and quizzes), submit assignments, take quizzes, view their own grades and performance reports, receive announcements and notifications, and communicate with teachers, but has no access to other users' data or system management functions. This diagram provides the view of the data exchange between the user's roles and the centralized system, ensuring a clear understanding of user boundaries, role-based access, and the system's scope as defined in the research objectives.

Figure 6

Data Flow Diagram Level 1 Super Admin



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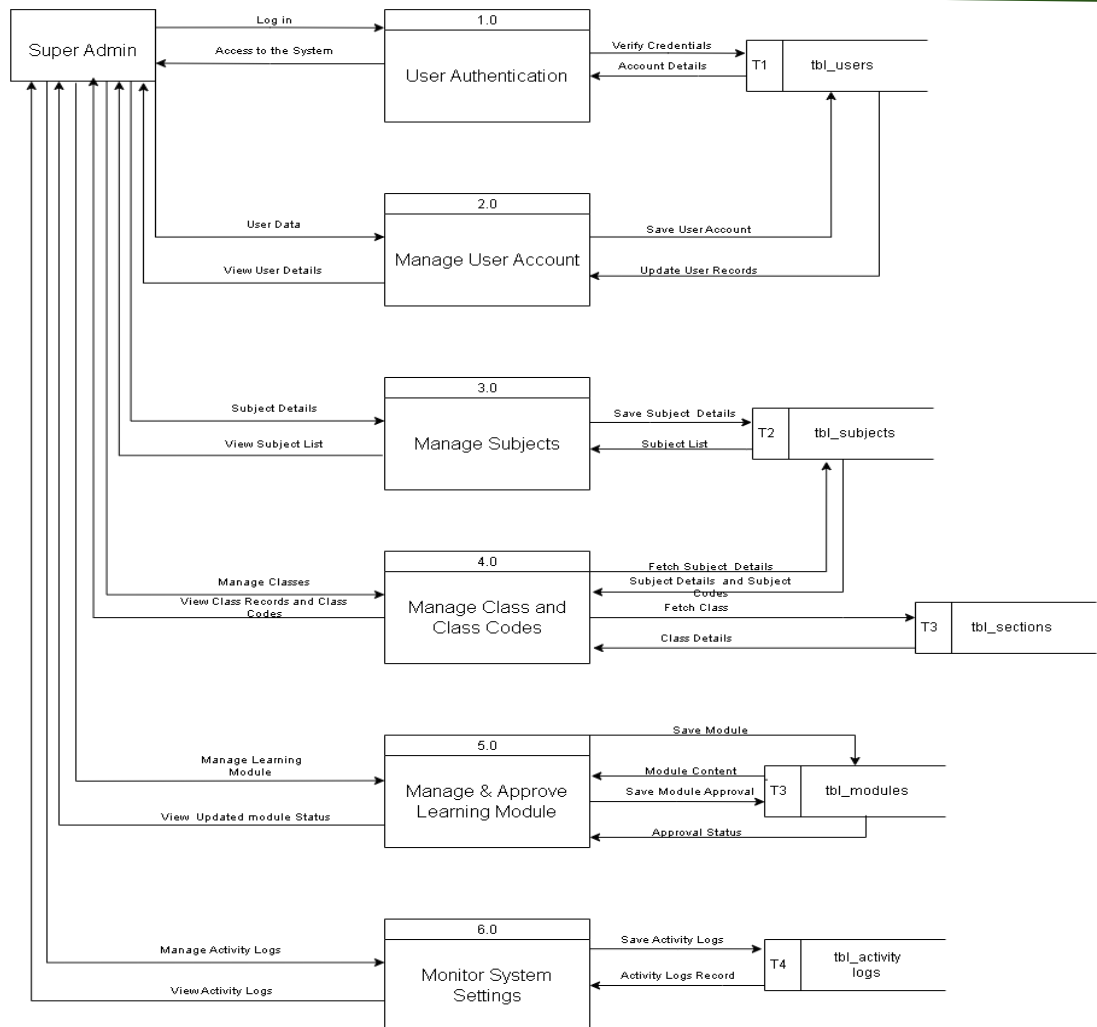


Figure 6 presents the data flow diagram for the *Super Admin module of the Interactive Learning Management System for CSS Students at Paliparan III Senior High School. This diagram decomposes the Super Admin's interactions into seven main processes, each connected to specific data stores that the system maintains. The first process is **User Authentication** (1.0) , where the Super Admin verifies credentials before gaining access to any administrative functions; this process reads from and writes to the users data store to validate login attempts and maintain



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session integrity. After authentication, the **Manage User Account** (2.0) process allows the Super Admin to create, edit, delete, and assign roles to all user accounts (students, teachers, and admins) by interacting with the users data store. The third process, **Manage Subjects** (3.0) , enables the Super Admin to add, modify, or remove subject offerings in the system, storing all subject- related data in the subjects data store. Next, **Manage Class and Class Codes** (4.0) gives the Super Admin the ability to create class sections, generate unique auto- enrollment codes for each class, and assign subjects to specific sections; this process uses the sections data store.

The fifth process, **Manage & Approve Learning Module** (5.0) , allows the Super Admin to review, approve, or reject learning modules submitted by teachers before they become visible to students; approved modules are stored in the modules data store. The sixth process, **Monitor System Logs** (6.0) , records all user activities (logins, file uploads, grade changes, and similar actions) into the activity log data store, and the Super Admin can query this data store to generate audit trails, security reports, and system health checks.

Figure # 6

Data Flow Diagram Level 1 Admin



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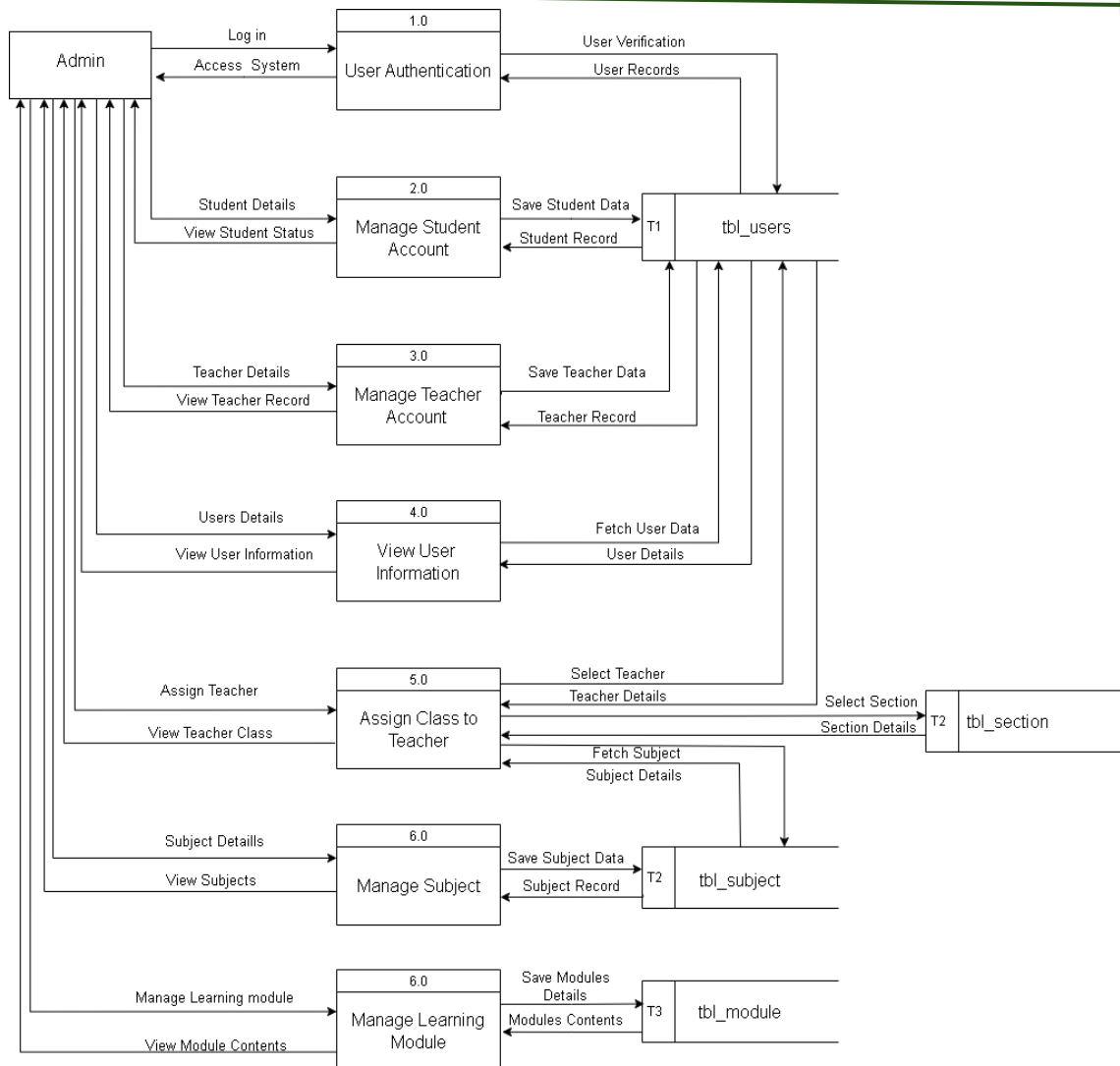


Figure 6 presents the data flow diagram for the Admin. This diagram decomposes the Admin's interactions into seven sequential processes, each connected to specific data stores that the system maintains. The first process is **User Authentication** (1.0) , where the Admin must verify login credentials before accessing any administrative functions; this process reads from and writes to the users data store to validate identity and maintain session security. After



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authentication, the **Manage Student Account** (2.0) process allows the Admin to create, edit, and delete student accounts, storing all student-related data in the users data store. Similarly, the **Manage Teacher Account** (3.0) process enables the Admin to create, edit, and delete teacher accounts, also interacting with the users data store. The fourth process, **View User Information** (4.0) , gives the Admin read- only access to view details of both student and teacher accounts (such as names, email addresses, and enrollment status) from the user's data store for monitoring and verification purposes. Next, **Assign Class to Teacher** (5.0) allows the Admin to link a teacher to a specific class section; this process reads from and writes to the sections data store to maintain accurate teacher- class assignments.

The sixth process, **Manage Subject** (6.0) , enables the Admin to add, modify, or remove subject offerings within the system, storing all subject data in the subjects data store. Finally, **Manage Learning Module** (7.0) gives the Admin the ability to review, approve, or reject learning modules submitted by teachers, as well as organize modules by subject and grade level; approved modules are stored in the modules data store. In summary, this data flow diagram illustrates that the Admin serves as an intermediate manager with responsibilities limited to user account management (students and teachers), class assignment, subject management, and module oversight, but without the Super Admin's access to system- wide settings, activity logs, or full configuration control, as defined in the research objectives.



Figure #6

Data Flow Diagram Level 1 Teacher

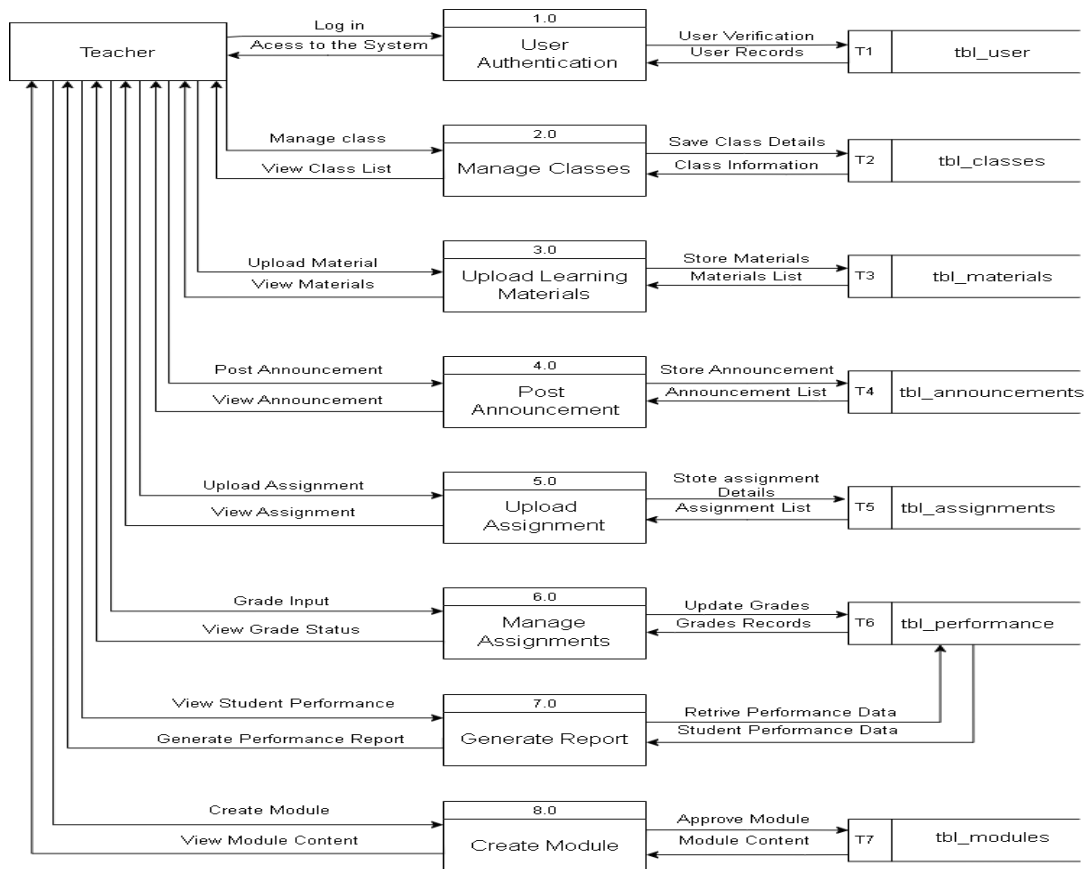


Figure 6 presents the data flow diagram for the Teacher. This diagram decomposes the Teacher's interactions into eight sequential processes, each connected to specific data stores. The first process is **User Authentication** (1.0) , where the Teacher verifies login credentials before accessing any teaching functions; this process reads from and writes to the users data store to validate identity and maintain session security. After authentication, the **Manage Classes** (2.0) process allows the Teacher to create classes with auto-generated enrollment codes, edit



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class details, and view enrolled students, storing all class- related data in the classes data store. The third process, **Upload Learning Materials** (3.0) , enables the Teacher to upload instructional resources such as PDFs, PowerPoint presentations, and videos, which are stored in the materials data store for student access. Next, **Post Announcement** (4.0) gives the Teacher the ability to send notifications and reminders to the entire class or individual students; these announcements are saved in the announcements data store. The fifth process, **Upload Assignment** (5.0) , allows the Teacher to create and post assignment instructions, attach files, set due dates, and assign point values, with all assignment details stored in the assignments data store.

Following that, **Manage Assignments** (6.0) enables the Teacher to view submitted student work, grade submissions (manually for essays or automatically for objective quizzes), provide feedback, and record scores; this process updates the performance data store to reflect student grades and completion status. The seventh process, **Generate Report** (7.0) , allows the Teacher to produce visual progress reports and analytics dashboards showing module completion rates, quiz results, assignment scores, and overall student performance, drawing data from multiple stores (performance, assignments, materials). Finally, **Create Module** (8.0) gives the Teacher the ability to design interactive learning modules that incorporate multimedia elements such as instructional videos, flashcards, and quizzes; these



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modules are stored in the modules data store and, depending on the system configuration, may require Super Admin approval before publication. In summary, this data flow diagram illustrates how the Teacher functions as the central content manager and academic facilitator, with full control over classes, learning materials, announcements, assignments, grading, reporting, and module creation, all while ensuring that each process is supported by a corresponding data store as defined in the research objectives.

Figure 7

Data Flow Diagram Level 1 Student



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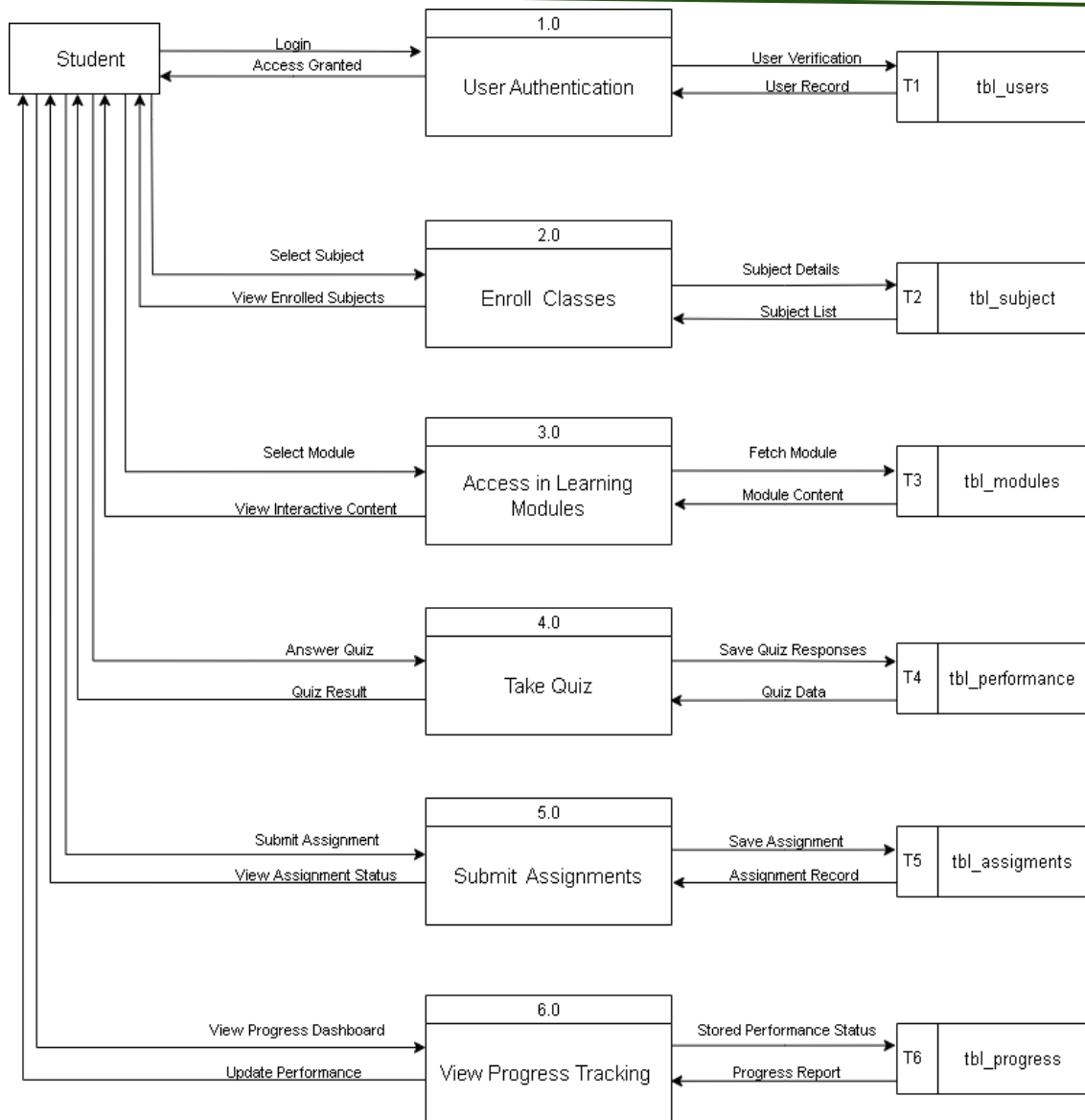


Figure 7 presents the data flow diagram for the Student. This diagram decomposes the Student's interactions into six sequential processes, each connected to specific data stores. The first process is **User Authentication (1.0)**, where the Student verifies login credentials before accessing any learning functions; this process reads from and writes to the users data store to validate identity and



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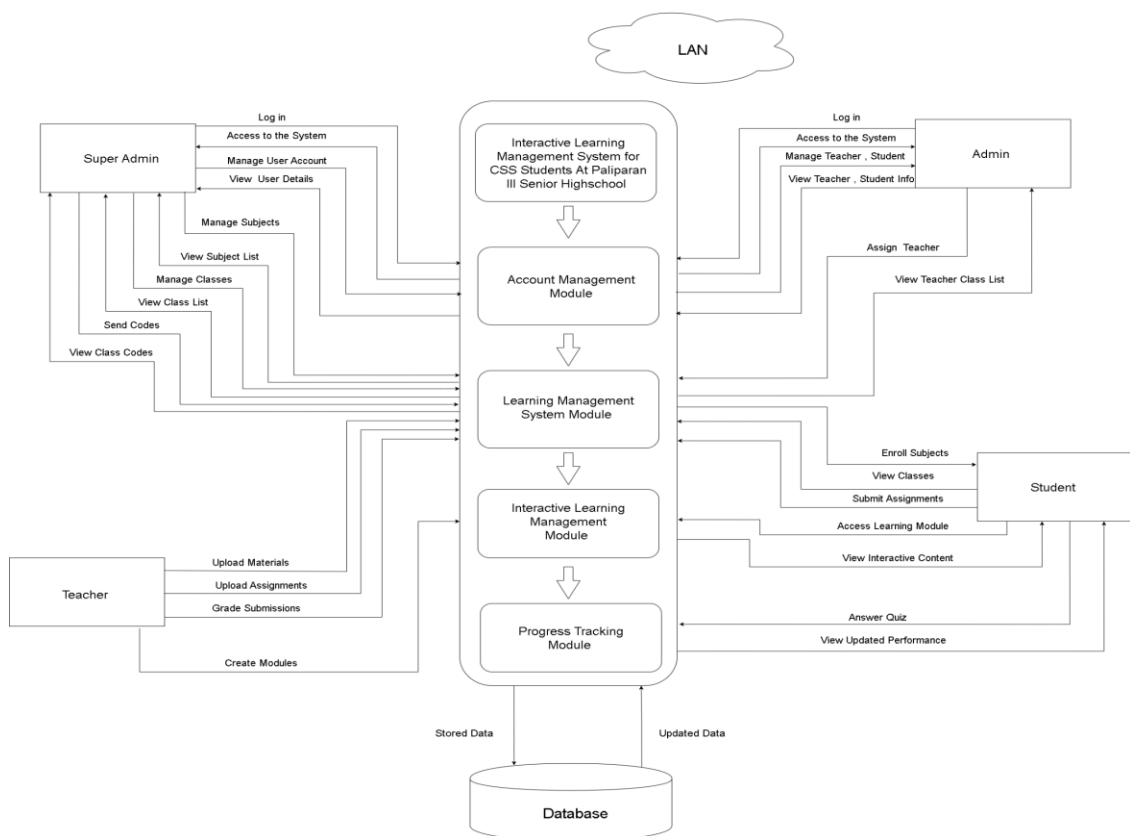
maintain session security. After authentication, the **Enroll Classes** (2.0) process allows the Student to join classes using unique auto-generated class codes provided by the teacher; this process reads from and writes to the subjects data store to record the Student's enrollment in specific subjects and sections. The third process, **Access Learning Modules** (3.0) , enables the Student to browse, view, and navigate through interactive learning modules that include multimedia content such as instructional videos, flashcards, and page- by- page lessons; all module content is stored in the modules data store and accessed by the Student as permitted by their enrolled classes. Next, **Take Quiz** (4.0) allows the Student to answer various types of quiz questions (multiple- choice, true/false, multi- select) that are part of the learning modules or assigned by the teacher; the Student's answers are automatically scored where applicable, and the results are recorded in the performance data store. The fifth process, **Submit Assignments** (5.0) , gives the student the ability to upload completed assignments (in PDF or other permitted formats) and submit them before the due date; submitted files are stored in the assignments data store for teacher access and grading. Finally, **View Progress Tracking** (6.0) allows the student to monitor their own academic performance through visual dashboards and reports, including module completion rates, quiz scores, assignment grades, and overall progress percentage; this information is retrieved from the progress data store and the performance data store. In summary, this data flow diagram illustrates how the Student functions as the primary learner, with capabilities to enroll in classes, access



interactive content, complete assessments, and track personal academic progress, all while ensuring that each process is supported by a corresponding data store as defined in the research objectives.

Figure 8

Theoretical Framework





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Figure 8 shows the theoretical framework of Interactive Learning Management System that illustrates the structural interaction among the users, modules, and data flow of the system. As shown in the diagram above, the framework is composed of four interconnected modules, the information flows vertically starting with the account management module where the users are authenticated, next is learning management module for managing lessons and activities down through interactive learning module for student's engagement and progress tracking module for monitoring performance. This vertical flow ensures that every action taken by the user is processed in a logical sequence before being stored and components are integrated to support the overall learning process. These modules are accessed by four primary users, the student, teacher, admin and super admin. Each user has specific roles and responsibilities that determine their relationship and how they interact with the system.

The process begins when users log into the system and gain access to its features. The super admin has the highest level of control in the system. It's also the one managing all the users accounts including admin, teacher and students, monitors overall system activity and oversees the learning management system and interactive learning module. As for admin, it is responsible for managing teacher and students accounts, this includes creating, updating and maintaining accounts for both students and teachers. The admin is also responsible for assigning subject, section



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and class groupings. In addition, the admin monitors and helps super admin to maintain the system by organizing records and ensuring that all users have access to the features. Through these tasks, the admin helps maintain functionality and efficiency of the system.

Teachers, on the other hand , manage the learning management system modules. This includes uploading learning materials, posting announcements, and providing other resources that students will use during the learning process. They can also upload assignments and assign grades to students based on the quality and accuracy of their work. They are also responsible for managing and recording the grades of submitted assignments within the platform. Additionally, teachers can create their own interactive learning modules such as activities , quizzes and exercises within the system. Although the system records student performance, the teacher primarily interacts with the LMS and can only access but cannot modify the Progress Tracking Module, as grading is handled within the platform.

As for the student user , after gaining access to the system they can now navigate through the learning management module where they can enroll subjects or class , view learning materials, and upload their assignments that they can submit directly to the platform. Students also can engage with the multimedia content in interactive learning modules such as videos , flashcards , activities , and quizzes to enhance their understanding. The system also automatically tracks their



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performance based on quiz results and module completion, and students can view their own progress reports and dashboard showing their strengths, weaknesses, and areas for improvement. However, students cannot modify any learning materials, grades, or system settings, their role is limited to learning, submitting work, and monitoring their own academic progress.

System Design

Forms and Reports

Figure 9

Login Form/ Sign Up

iLearn

Let's the learning journey begin.

Unlock a world of learning with just one click. Log in to get started.

Username or Email

Enter username or email

Password [Forgot Password?](#)

Enter password

Login

Don't have an account? [Sign up here to create](#)

Need help? Contact us at helloilearn@gmail.com



Figure 10

Sign up Form

iLearn

Let's the learning journey begin.

Unlock a world of learning with just one click. Log in to get started.

Student ID / LRN

e.g. 123456789012

Firstname M.I Lastname

e.g. Juan e.g. D e.g. Dela Cruz

Email

e.g. juandelacruz@gmail.com

Username

e.g. juan_dc12

Username

e.g. juan_dc12

Password

At least 8 characters

Confirm Password

Re-enter your password

Grade Level Section

Select grade level Select section

Signup

Already have an account? [Sign in here to proceed](#)

Need help? Contact us at helloilearn@gmail.com

Figure 11

Student Dashboard

Figure 12



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Teacher Dashboard

Figure 13

Admin Dashboard

Figure 14

Super Admin Dashboard

Figure 15

Assignment Dashboard

Figure 16

Progress Report

Figure 17

Report Dashboard



Figure

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Entity

Relationship

Diagram

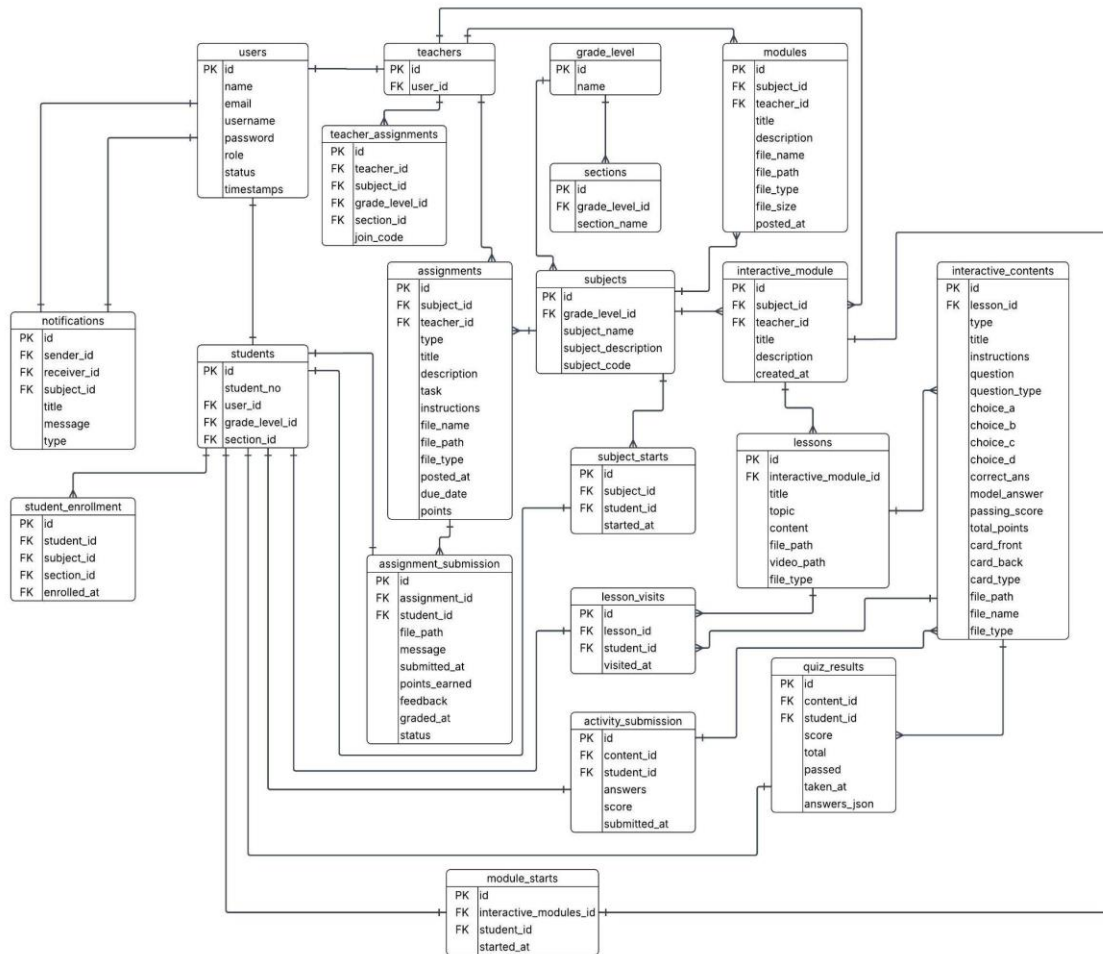


Figure 18 shows the Entity Relationship Diagram (ERD) of the Interactive Learning Management System. It is a visual database modeling tool that represents the entities (data objects or concepts) within a system and the relationships between them. Entities are shown as rectangles and contain specific attributes that describe them, such as a user's name, email, and role, or a module's title, file path, and posted date. Relationships between entities define how they interact, such as a teacher



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uploading multiple modules or a student submitting many assignments. Standardized symbols include primary keys (PK), foreign keys (FK), and cardinality indicators (one- to- many, one- to- one) that specify how many instances of one entity relate to another.

Data Dictionary

Table 1

users

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
name	varchar	255	NULL	Name of the user
email	varchar	255	NULL	Email of the user
username	varchar	255	NULL	Username of the user
password	varchar	255	NULL	Password of the user
role	varchar	255	NULL	Role of the user
status	varchar	255	NULL	Status of the user
created_at	timestamp	-	AUTO	Record creation date
updated_at	timestamp	-	AUTO	Record update date



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students

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
student_no	varchar	255	NOT NULL	Student number of user
user_id	int	11	FK	Foreign key user id
grade_level_id	int	11	FK	Foreign key grade level id
section_id	int	11	FK	Foreign key section id

subjects

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
grade_level_id	int	11	FK	Reference id in the grade_level table. identifies the grade level of the students
subject_name	varchar	255	NOT NULL	Name of the subject
subject_description	varchar	255	NOT NULL	Description of the



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				subject
subject_image	varchar	255	NOT NULL	Image of the subject
subject_code	varchar	255	NOT NULL	Code in the url of the subject

teacher_assignments

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
teacher_id	int	11	FK	Reference id in the teachers table identifies the teachers records
subject_id	int	11	FK	Reference id in the subjects table. identifies the subjects
grade_level_id	int	11	FK	Reference id in the grade_level table. identifies the grade level of the student



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section_id	int	11	FK	Reference id in the sections table. identifies the section of the student
join_code	varchar	255	NOT NULL	Class code of the subject

sections

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
grade_level_id	int	11	FK	Reference id in the grade level table. identifies the grade level of the student
section_name	varchar	255	NOT NULL	Refer to the section of the student

teachers

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
user_id	int	11	FK	Reference id in the users



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				table. identifies the user associated with the record
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student_enrollments

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
student_id	int	11	FK	Reference id in the students table. identifies the students
subject_id	int	11	FK	Reference id in the subjects table. identifies the subjects
section_id	int	11	FK	Reference id in the sections table. identifies the sections of the students
enrolled_at	datetime	-	NOT NULL	Refer to the date and time that they enroll



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subject_starts

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
subject_id	int	11	FK	Reference id in the subjects table. identifies the subjects
student_id	int	11	FK	Reference id in the students table. identifies the students
started_at	datetime	-	NOT NULL	Refer the date and time that the student start the subject

quiz_results

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
content_id	int	11	FK	Reference id in the interactive_c contents table. identifies the content of



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				the quiz
student_id	int	11	FK	Reference id in the students table. identifies the students
score	int	11	NOT NULL	Refer to the score of the students that got in the quiz
total	int	11	NOT NULL	Refer to the possible total score in the quiz
passed	tinyint	1	NULL	Indicates if the student passed or not
taken_at	timestamp	-	NOT NULL	Indicates the date and time when the quiz was taken
answers_json	text	-	NULL	Stores the student's answers in JSON

notifications

Field name	Data type	Field length	Constraints	Description
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id	int	11	PK, Auto increment	Unique user id
sender_id	int	11	FK	Reference id in the users table. identifies the user
receiver_id	int	11	FK, NULL	Reference id in the users table. identifies the user
subject_id	int	11	FK, NULL	Reference id in the subjects table. identifies the subject
title	varchar	255	NULL	Refer to the title of the announcement
message	text	-	NOT NULL	Refer to the message of the teachers and students
type	varchar	255	NOT NULL	Refer to the type if is it announcements or messages
created_at	timestamp	created_at	NOT NULL	Indicates the time when the announcement



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				nts or messages created
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module_starts

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
interactive_modules_id	int	11	FK	Reference id in the interactive_modules table. identifies the interactive modules content
student_id	int	11	FK	Reference id in the students table. identifies the students
started_at	datetime	-	NULL	Indicates the date and time when the modules starts

modules

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id



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subject_id	int	11	FK	Reference id in the subjects table. identifies the subjects
teacher_id	int	11	FK, NULL	Reference id in the teachers table identifies the teachers records
title	varchar	255	NOT NULL	Refer to the title of the modules
description	text	-	NULL	Refer to the description of the module
file_name	varchar	255	NULL	Refer to the file name of the pdf file
file_path	varchar	255	NULL	Refer to the file path of the pdf file
file_type	enum	-	NULL	Refer to the what file type of the pdf file
file_size	int	11	NULL	Refer to the what size of the pdf file



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posted_at	timestamp	-	NOT NULL	Indicates the time when the modules is created
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lesson_visits

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
lesson_id	int	11	FK	Reference id in the lessons table identifies the lessons
student_id	int	11	FK	Reference id in the students table. identifies the students
visited_at	datetime	-	NULL	Indicates the date and time when the lessons visit or open

lessons

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id



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interactive_module_id	int	11	FK	Reference id in the interactive_modules table identifies the interactive_modules content
title	varchar	255	NOT NULL	Refer to the title of the lessons
topic	varchar	255	NULL	Refer to the topic of the lessons
content	longtext	-	NULL	Refer to the content of the lessons
file_path	text	-	NULL	Refer to the file path of the pdf file
video_path	text	-	NULL	Refer to the video path of the pdf file
file_type	varchar	20	NULL	Refer to the file type of the pdf file

interactive_modules

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
subject_id	int	11	FK	Reference id



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				in the subjects table. identifies the subjects
teacher_id	int	11	FK	Reference id in the teachers table identifies the teacher who creates the interactive_modules
title	varchar	255	NOT NULL	Refer to the title of the interactive_modules
description	text	-	NULL	Refer to the description of the interactive_modules
created_at	timestamp	-	NOT NULL	Indicates the time when the interactive_modules was created

interactive_contents

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id



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lesson_id	int	11	FK, NULL	Reference id in the lessons table. identifies the lessons
type	enum	('quiz', 'activity', 'flashcard', 'video', 'image')	NULL	Refer to the what type of interactive_c ontents
title	varchar	255	NULL	Refer to the title of the interactive_c ontents
instructions	text	-	NULL	Refer to the instructions of the interactive_c ontents
question	text	-	NULL	Refer to the questions of the interactive_c ontents
question_type	varchar	50	NULL	Refer to the what question type if choose the activity
choice_a	varchar	255	NULL	Refer to the choices letter a in the quiz



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choice_b	varchar	255	NULL	Refer to the choices letter b in the quiz
choice_c	varchar	255	NULL	Refer to the choices letter c in the quiz
choice_d	varchar	255	NULL	Refer to the choices letter d in the quiz
correct_ans	varchar	10	NULL	Refer to the correct answer of the quiz
model_answer	text	-	NULL	Refer to the model answer of the activity
passing_score	int	11	NULL	Refer to the passing score of the quiz and activity
total_points	int	11	NULL	Refer to the total points of the activity
card_front	text	-	NULL	Refer of the flash card front card content
card_back	text	-	NULL	Refer of the flash card back card content



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card_type	varchar	50	NULL	Refer what type of card is flash card
file_path	varchar	255	NULL	Refer to the file type of the image or videos
file_name	varchar	255	NULL	Refer to the name of the file pdf
file_type	varchar	50	NULL	Refer what file type is the file chosen
created_at	timestamp	-	NOT NULL	Indicates the time when the interactive_c contents created
grade_level				
Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
name	varchar	255	NOT NULL	Refer to the name what grade level is it like Grade 12 and Grade 11



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assignment_submissions

Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
assignment_id	int	11	FK	Reference id in the assignments table. Identifies the student who passed the assignment
student_id	int	11	FK	Reference id in the students table. Identifies the students who submit the assignment
file_path	varchar	500	NULL	Refer to the file path of the pdf file
file_name	text	-	NULL	Refer to the file name of the pdf file
submitted_at	timestamp	-	NOT NULL	Refer to the time of submitted
points_earned	int	11	NULL	Refer to the what points that the students



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				earned or grade
feedback	text	-	NULL	Refer to the feedback of the teacher to the assignment that submitted
graded_at	datetime	-	NULL	Refer to the date and time that the assignment was graded
status	enum	('submitted', 'late', 'missing')	NULL	Refer of the assignment's status
assignments				
Field name	Data type	Field length	Constraints	Description
id	int	11	PK, Auto increment	Unique user id
subject_id	int	11	FK, NULL	Reference id in the subjects table. Identifies the what subjects is the assignment will upload
teacher_id	int	11	FK, NULL	Reference id in the



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				teachers table. Identifies who the teacher will upload the assignment
type	varchar	255	NULL	Refer to what type of assignments is it
title	varchar	255	NOT NULL	Refer to the title of the assignment
description	text	-	NULL	Refer to the description of the assignment
task	text	-	NULL	Refer to what task is it
instructions	text	-	NULL	Refer to the instructions that they needed to check first before answering the assignment
file_name	varchar	255	NULL	Refer to the file name of the pdf that in the assignment



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				included
file_path	varchar	255	NULL	Refer to the file path of the file that uploaded
file_type	enum	('pdf')	NULL	Refer to what file type is it
posted_at	timestamp	timestamp	NOT NULL	Refer to what time that the assignment uploaded
due_date	date	date	NULL	Refer to the due date that the assignment needed to pass
points	int	11	NULL	Refer to the points that they need to get
created_at	datetime	-	NULL	Indicates to what date and time that the assignment was created

activity_submissions

Field name	Data type	Field length	Constraints	Description
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id	int	11	PK, Auto increment	Unique user id
content_id	int	11	FK	Reference id in the interactive_c ontents
student_id	int	11	FK	Reference id in the students table. Identifies the students who submit the activity
answers	longtext	-	NULL	Refer to the answers of the students
score	int	11	NOT NULL	Refer to the score of the students in activity
submitted_at	timestamp	-	NOT NULL	Refer to the time when the activity was submit

System Architecture

Figure 19

Network Model



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The proposed system follows a client-server network model, where multiple users such as students, teachers, super admin and administrators act as clients that access the system through computers or mobile devices. The central server in this architecture stores the database and all essential system components which include Account Module, LMS, Interactive Learning Module, and Progress Tracking Module, acting as the 'Provider.' The system design enables centralized data management, ensuring that all information such as user accounts, learning materials, and student performance records are stored and maintained in a single database. The system provides users with an effective communication tool which functions throughout the school setting.

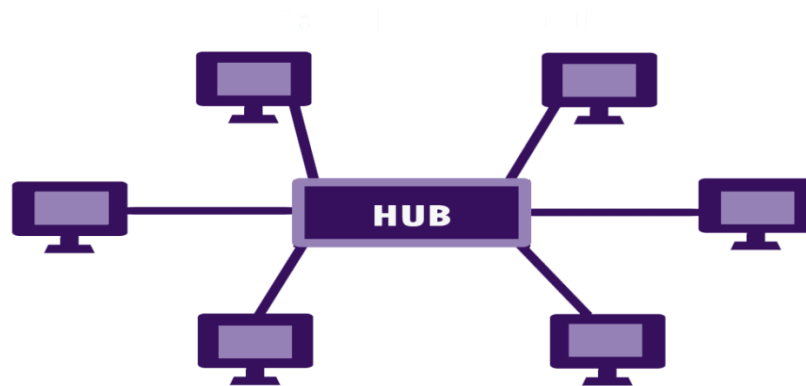
Figure 20

Star Hub Network Topology



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The proposed system uses a star topology to build its Local Area Network (LAN) at Paliparan III Senior High School. In this design, every client device including student mobile phones, teacher computers, and admin terminals connects directly to a central network switch. The central switch then links to the application server and database server, where all system modules and user data are stored. Star topology was chosen primarily because it offers high fault tolerance. The system maintains uninterrupted operation when users such as a student or teacher loses phone connection and experiences computer cable failure. For a school environment, this reliability is critical because students need to access learning materials and quizzes continuously during class hours. The central switch enables IT staff to determine which cable or port is malfunctioning when they investigate a network problem that has occurred. The targeted approach resolves the problem because it allows the system to function without interrupting network operations or disturbing users. Finally, star topology supports scalability, allowing the school to add more



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devices such as new student phones or teacher computers by simply plugging them into available switch ports. As the number of CSS students grows, this flexibility ensures the network can expand without costly redesign or major reconfiguration.

Security

To protect user data and maintain system integrity, the Interactive Learning Management System implements security measures tailored to its academic environment. User authentication is enforced through password hashing and secure login sessions, ensuring that only registered students, teachers, admins, and the super admin can access the platform. Role- based access control restricts what each user can do: students can only view their own enrolled subjects, modules, and grades; teachers can upload materials, create quizzes, grade submissions, and see only their assigned classes; admins can manage student and teacher accounts but cannot alter system settings; and the super admin has full privileges to configure the system, approve modules, and view all activity logs. All sensitive data, including student grades, assignment submissions, and quiz results, are stored in an encrypted database (MySQL) with regular backups to external and cloud storage. The system also maintains detailed audit logs (tbl_activity) that record every critical action such as module uploads, grade changes, class creation, and account modifications, providing a tamper- evident trail for accountability. Communication between the



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user's browser and the web server is secured via HTTPS with SSL/TLS encryption, preventing interception of login credentials or submitted assignments

Table 2

Software Specification

Components	Specification	Description
Development OS	Windows 10 / Linux	Platform for coding, testing, and debugging the LMS
Web Server	Apache (via XAMPP)	Hosts the LMS and handles client HTTP requests
Programming Language		
Database Management System	MySQL 8.0	Stores user accounts, subjects, classes, modules, assignments, submissions, grades, logs, and settings
Integrated Development Environment (IDE)	Visual Studio Code	Primary code editor for developing and debugging the LMS
Local Development Environment	XAMPP (Apache + MySQL + PHP)	Provides offline local server for testing before deployment
Web Browser (Client Access)	Google Chrome / Mozilla Firefox / Microsoft Edge	Allows students, teachers, and admins to access the LMS via standard browsers

Table 2 outlines the software requirements for the Interactive Learning Management System for CSS Students at Paliparan III Senior High School. The components include development tools (Visual Studio Code), server technologies (Apache, PHP), database management (MySQL 8.0), local testing (XAMPP), and client browsers (Chrome, Firefox, Edge). PHP and MySQL enable automated progress tracking, interactive modules, quiz generation, and real-time dashboards. Apache



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ensures stable handling of concurrent user sessions, while XAMPP allows offline testing. Modern browsers guarantee cross- platform compatibility for all end users, supporting the system’s goals of improving learning efficiency, engagement, and performance monitoring for CSS students.

Table 3

Hardware Specification

Components	Minimum Specification	Recommended Specification
Device	Laptop or Desktop	Laptop + Desktop
Processor	Intel Core i3 or equivalent	Intel Core i5 or higher
RAM	4 GB	8 GB or higher
Storage	128 GB HDD	256 GB SSD or higher
Network	Stable internet (5 Mbps)	High-speed internet (10–20 Mbps)

Table 3 presents the hardware and system requirements for the Interactive Learning Management System for CSS Students at Paliparan III Senior High School, outlining minimum and recommended specifications across six components: device, processor, RAM, storage, network, and backup. For the device, a laptop or desktop is the minimum to allow students and teachers to access the platform, while the recommended laptop and desktop combination supports productivity for admins



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and super admin who manage user accounts, subject offerings, and system settings. An Intel Core i3 processor with 4 GB RAM meets basic needs such as browsing modules and taking quizzes, but an i5 processor with 8 GB RAM or higher is recommended for teachers generating performance reports and for the super admin monitoring activity logs and configuring system parameters. A 128 GB HDD is sufficient for storing core system files, whereas a 256 GB SSD improves load times for interactive multimedia modules and dashboards. A stable 5 Mbps internet connection enables submission of assignments and access to learning materials, while 10–20 Mbps ensures smooth streaming of instructional videos, real-time quiz feedback, and handling of multiple concurrent users as specified in the system's objectives. Finally, external flash drive backup serves as a minimum for manual data saving, but cloud backup combined with external storage is recommended to guarantee data redundancy, disaster recovery, and long-term integrity of student progress records, grades, and activity logs.

Program Specification

The Interactive Learning Management System (iLearnSystem) is a web-based platform designed for the CSS strand of Paliparan III Senior High School to replace



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manual and fragmented learning processes. It supports four user roles—Super Admin, Admin, Teacher, and Student—each with specific responsibilities, enabling efficient management of learning materials, assessments, communication, and student performance tracking.

The system collects structured inputs such as user account data, subject details, instructional materials, quizzes, assignments, and student submissions. These inputs are processed through a PHP and MySQL-based client-server system organized into four main modules: Account Management, LMS, Interactive Learning, and Progress Tracking. These modules handle authentication, content delivery, interactive activities, and performance analysis while ensuring data integrity and controlled access.

Outputs include interactive learning content, dashboards, automated quiz results, progress reports for students, and performance analytics for teachers and administrators. The system integrates features like role-based access, assignment management, automated grading, communication tools, and audit logs, all connected through a centralized database.

The platform is built using standard web technologies (HTML, CSS, JavaScript, PHP, MySQL) and runs on typical computer hardware with internet access. It undergoes comprehensive testing, including unit, integration, system, performance, and regression testing, to ensure reliability and efficiency.



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Finally, the system is evaluated using the ISO 25010 quality standards, focusing on functionality, performance, usability, security, reliability, and portability. A numerical rating scale is used to assess overall system effectiveness and identify areas for improvement.

Programming Environment

For the frontend, the system utilizes HTML for structuring web pages, CSS and Bootstrap for responsive and consistent styling across different screen sizes (desktops, laptops, tablets), and JavaScript for client-side interactivity such as real-time form validation, dynamic content loading, and interactive quiz elements. This frontend ensures that students can easily navigate learning modules, teachers can manage classes through intuitive dashboards, and administrators can access all features without performance lag. For the backend, the system is built with PHP to handle server-side logic including user authentication, role-based access control, module upload and retrieval, quiz scoring, assignment submission processing, automated progress tracking, and report generation. MySQL (SQL) serves as the database management system, storing all persistent data such as user accounts (students, teachers, admin, super admin), subjects, class sections, enrollment codes, learning modules, assignments, quiz questions, submissions, grades, progress records, activity logs, and system settings. The choice of PHP and MySQL aligns with



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the system's need for a cost- effective, widely supported, and secure environment that can be hosted on a standard web server

Test Plan

To thoroughly evaluate First, to verify functionality, the researcher will ensure that all system features operate as specified, including user authentication and role- based access (student, teacher, admin, super admin), class creation with auto- generated codes, module upload and approval, quiz creation with automatic scoring, assignment submission and grading, progress tracking dashboards, announcements and notifications, calendar functions, and communication messaging. Second, to validate security, the researcher will test authentication protocols (login credentials, session management), authorization rules (role- based restrictions ensuring students cannot access teacher functions, etc.), data encryption for sensitive information, and audit logs for all user activities as defined in the security and data integrity objective. Third, to test performance, the researcher will measure system response times under varying load conditions, simulating multiple concurrent users (students taking quizzes, teachers uploading modules, super admin generating reports) to identify potential bottlenecks in the web server (Apache) and database (MySQL).



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Fourth, to verify reliability, the researcher will test backup and recovery mechanisms (external flash drive and cloud backup) to guarantee that student progress records, grades, submissions, and system logs can be restored effectively in the event of hardware or software failure. Fifth, to evaluate the system environment, the researcher will test web server configuration (Apache), PHP 8.x compatibility, MySQL query performance, browser rendering consistency (Chrome, Firefox, Edge), and network stability under real-world deployment conditions to ensure seamless operation for students and teachers.

System Testing Procedure

The testing procedure for the iLearnSystem is designed to systematically verify that all functional and non-functional requirements identified during the development phase are correctly implemented and operate as specified. The procedure encompasses five types of testing: functional testing, security testing, performance testing, reliability testing, and environment and compatibility testing. Each type is conducted in a structured sequence, progressing from unit-level verification to integrated system-wide evaluation.

Table 4



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Table 1: Unit Testing for Interactive Learning Management System (iLearnSystem)

Test Area	Objective	Tools	Responsible Tester
Login Module	Verify correct login and validation	Browser, XAMPP	Developer
User Management	Ensure CRUD operations work	PHPMyAdmin	Developer
Quiz Module	Check scoring and responses	Browser	Developer

Table 4 presents the Unit Testing conducted for the iLearnSystem. This table identifies the specific modules tested individually, including login, user management, quiz, and assignment modules. The objective column describes the purpose of each test, while the tools column indicates the software used. The responsible tester column identifies the assigned personnel. This ensures each component functions correctly before integration.

Table 5



Table 2: Integration Testing for Interactive Learning Management System (iLearnSystem)

Test Area	Objective	Tools	Responsible Tester
Login → Dashboard	Ensure correct redirection by role	Browser	Developer
LMS + Interactive Module	Verify content loads correctly	Browser	Developer
Quiz + Database	Ensure answers are stored and retrieved	MySQL	Developer
Assignment + Grading	Check submission to grading flow	Browser	Developer

Table 5 shows the Integration Testing for the iLearnSystem. It focuses on how different modules interact, ensuring proper data flow and communication between components. The objectives validate each integration point, while tools and testers ensure accurate execution. This guarantees that combined modules function as one system.

Table 6



Table 3: System Testing for Interactive Learning Management System (iLearnSystem)

Test Area	Objective	Tools	Responsible Tester
Full System	Verify end-to-end functionality	Browser	QA Tester
Workflow			
User Roles	Ensure role-based access works	Browser	QA Tester
Dashboard Display	Check correct data visualization	Browser	QA Tester
Notifications	Verify announcements and alerts	Browser	QA Tester

Table 6 presents the System Testing for the iLearnSystem. It evaluates the complete system including workflows, roles, dashboards, and notifications. The objectives ensure all features operate correctly in real scenarios. Tools and testers simulate user interaction to confirm system reliability.

Table 7



Table 4: Performance Testing for Interactive Learning Management System (iLearnSystem)

Test Area	Objective	Tools	Responsible Tester
Multiple Users	Test system under concurrent access	Browser	QA Tester
Dashboard Loading	Measure load time	DevTools	QA Tester
Quiz Submission	Check response time	Browser	QA Tester
File Upload	Test speed of uploads	Browser	QA Tester

Table 7 illustrates the Performance Testing for the iLearnSystem. It evaluates responsiveness, speed, and stability under load. The objectives measure system efficiency, while tools monitor performance. This ensures the system performs well even with multiple users.

System Evaluation Procedure

The evaluation of Interactive Learning Management System (iLearnSystem) will be based on the ISO 25010 standard, which includes key quality characteristics: functional suitability to assess if the system meets the required tasks like Learning management and Progress Tracking , performance efficiency to measure the system's responsiveness under peak loads, compatibility to ensure seamless integration with



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other systems, usability to evaluate the ease of use for both teacher and students, reliability to ensure consistent performance without failures, security to protect sensitive data and ensure compliance with privacy regulations, maintainability to allow for easy updates and modifications, and portability to ensure the system can operate across different environments. These criteria will ensure the system meets the quality standards needed for optimal performance and user experience.

Standard Numerical Scale for System Evaluation

Table 8

Numerical Scale Table

NUMERICAL SCALE	INTERPRETATION
4.21 – 5.00	Excellent
3.41 – 4.20	Very Good
2.61 – 3.40	Good
1.81 – 2.60	Fair
1.00 – 1.80	Poor

Table 8 presents the numerical scale that will be used to evaluate the Interactive Learning Management System for CSS Students at Paliparan III Senior High School. The scale translates numerical ratings from 1.00 to 5.00 into qualitative interpretations ranging from Poor to Excellent. This structured approach ensures that feedback collected from students, teachers, IT professionals, and non- technical users is analyzed objectively based on the ISO 25010 software quality characteristics



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(functionality, reliability, usability, efficiency, security, maintainability, and portability). The evaluation results will determine whether the system successfully meets its objectives of providing interactive learning modules, automated progress tracking, real- time dashboards, and efficient communication between teachers and students.